Selection Beats Compression: Memory Decision Policies for LLM Agents Under a Bounded Store

Simar Singh Khanuja
Dept. of Computer Science and Engineering
SVKM's NMIMS, Indore Campus
Indore, India

Yash Ramchandani
Dept. of Computer Science and Engineering
SVKM's NMIMS, Indore Campus
Indore, India

***Abstract*—A long-running LLM agent cannot keep every turn of a conversation in its context window, so a memory layer has to decide what to keep, what to compress, what to forget and what to recall. Tiered memory systems that make these decisions are already in use, but the decision policy itself has rarely been isolated and measured. This paper presents MemGate, an evaluation harness that places competing memory policies on a common accuracy-per-token frontier. Each comparison is set up so that only one factor differs between the two arms. We first show that the accounting normally used in this area, which bounds the context assembled per query but leaves the store unbounded, charges a store-everything baseline nothing for retention and therefore settles the comparison in advance. When the store is also bounded, choosing which turns to forget is worth $+9.37$ answer-recall points over forgetting oldest-first at identical storage, fidelity and retrieval (95% CI $[+5.45, +14.90]$ clustered by conversation, McNemar $p = 2.8{\times}10^{-5}$), while compressing an evicted turn instead of dropping it loses 7.13 points. We also test the rate–distortion prediction that compression must eventually win. Up to a compression ratio of $182\times$ it does not, although the margin in favour of selection narrows. Charging the budget for the embedding index, which is normally left out, reverses the ranking below about 40 KiB, where a policy that stores no index at all performs best. Finally, an LLM-judged salience scorer performs worse than a regular-expression heuristic by 5.87 points, with the confidence interval excluding zero, whereas a logistic scorer over shallow features and a sentence embedding, trained on LoCoMo and on REALTALK conversations between real people, beats the heuristic by $+14.55$ points on held-out LoCoMo conversations (95% CI $[+10.40, +19.70]$). Across store budgets from 2048 to 16384 tokens its LoCoMo gain stays significant, and on REALTALK it becomes significant once the store holds about 40% of a conversation. Through a fixed local reader over all 715 answer-recoverable questions, $+6.0$ points of the selection advantage convert into correct answers (95% CI $[+3.3, +9.1]$), which is 69% of the recall gap in that run. We release the policy as a working chat application that runs offline on a laptop, and report three measurement errors that changed our own results, each found by an invariant rather than by inspection.**

***Index Terms*—large language models, agent memory, context management, retrieval-augmented generation, rate–distortion, empirical evaluation**

I. Introduction

An agent that runs for weeks accumulates a history far larger than any context window, so something must decide what enters the prompt each turn. A longer window does not remove the problem: accuracy falls 30–50% well before the documented limit across frontier models [1], and position inside a long context changes how much the model uses [2]. These are properties of attention over long inputs, not gaps in training, so a larger window moves the difficulty rather than solving it. The limiting factor is curation, not capacity.

The cost of poor curation can be measured. On LongMemEval, oracle retrieval scores about 92% where the same model retrieving for itself scores about 58% [3]. The model answers correctly once the right evidence is placed in front of it, so this 34-point difference is a selection failure rather than a reasoning failure.

Systems that address selection already exist. Tiered memory with summarisation and forgetting is used in MemGPT [4], MemoryBank [5] and the architectures reviewed in [6]. What has not been established is whether the individual mechanisms inside those systems earn their cost, since they are evaluated as complete systems rather than as separate decisions. A recent survey states the gap directly: current systems "do not solve the fundamental challenge: deciding what to remember and what to forget" [6]. A rate–distortion framing has been proposed for this family of decisions [7], but it provides an analytical framework rather than a measured system.

This paper measures the decision. Our research question is: how much of the memory-selection gap can a better decision policy recover, and under what budget does compression become worthwhile?

Our contributions are as follows. (1) A storage-budgeted protocol correcting an accounting error common in this area: bounding only the assembled context charges a store-everything baseline nothing for retention, settling the comparison before it is run (Section VI-A). (2) Evidence that the gain is selection, not compression — at identical storage, fidelity and retrieval, scored forgetting is worth $+9.37$ answer-recall points over FIFO while compressing an evicted turn loses 7.13 (Sections VI-B and VI-C). (3) A test of the rate–distortion crossover, to $182\times$ compression (Section VI-D). (4) Byte-denominated accounting: the embedding index outweighs the text it indexes by about $12\times$ and reverses the policy ranking below roughly 40 KiB (Section VI-E). (5) A negative result for LLM-judged salience — as a single-component swap, a 0.5B judge loses to a hand-written heuristic by 5.87 points (Section VI-F). (6) A learned scorer trained on two corpora, one LLM-generated and one written by people, with the direction in which it transfers and the direction in which it does

not (Section VI-G). (7) An end-task evaluation through a fixed reader, measuring how much of the recall advantage becomes correct answers (Section VI-H). (8) A working application built on the same code paths as the experiments (Section VII). (9) Three measurement errors in our own results, with the checks that caught them (Section VIII).

Four of these nine results are negative or corrective. We consider this a useful outcome. Described only as a three-tier memory layer, the system would duplicate work that has already been done; measured against controls, it indicates which parts of that design actually pay for themselves.

II. Related Work

MemGPT [4] pages a virtual context against external storage under the model's own control; MemoryBank [5] adds an Ebbinghaus forgetting curve; Generative Agents [8] score memories by recency, importance and relevance; A-MEM [9] organises them as interlinked notes. This work shows tiering and forgetting are practical, but because these systems are evaluated end to end, the contribution of any single decision inside them cannot be separated from the pipeline. Our harness is built to make that separation possible.

A second line compresses the prompt or the attention cache rather than choosing between stored items: LLMLingua [10] prunes tokens by perplexity, Gist tokens [11] and Auto-Compressors [12] learn dense summaries, and StreamingLLM [13] and H_2O [14] evict from the KV cache by attention-derived scores. All operate within a single forward pass over a contiguous context. The decision studied here is upstream and persistent: it fixes which items survive across sessions, where an evicted item cannot be recovered from the cache at all.

LoCoMo [15] supplies multi-session dialogues annotated with gold evidence turn identifiers, which is what lets a retrieval-level metric be computed without a generative model; LongMemEval [3] isolates several memory abilities and is the source of the gap quoted above. REALTALK [16] applies LoCoMo's annotation schema to 21 days of messaging between real people, which makes it the natural test of whether a scorer fitted on LoCoMo learned what matters or only what LLM-generated dialogue looks like. We propose neither a new architecture nor a new benchmark: the contribution is a measurement protocol and its results, including a case where correcting the protocol reverses the conclusion.

III. The MemGate Harness

MemGate is placed between an agent and a frozen LLM, and controls only what enters the prompt. No component is trained and the base model is treated as a black box, so any difference we report can be attributed to the memory policy.

A. Tiers as a lifecycle

The store has three tiers, but they are organised as a lifecycle rather than as a partition. Every turn enters Tier 1 verbatim, and the routing decision is taken once, at eviction, when the budget binds. Writing u for the utility assigned to an item by the scorer:

$$\begin{array}{ll} u \geq \tau_{\text{fact}} & \rightarrow \text{ Tier 3: atomic fact + vector} \\ u \geq \tau_{\text{low}} & \rightarrow \text{ Tier 2: compressed gist} \\ \text{otherwise} & \rightarrow \text{ dropped} \end{array}$$

Tier 1 holds recent verbatim turns and answers the question *when*; Tier 2 holds compressed gists and answers *gist*; Tier 3 holds durable facts with embeddings, answers *what*, and enters the context only when retrieval matches. Scoring each item once, at eviction, is what makes the decision point separable, since it gives a single site in the pipeline that a controlled swap can target.

B. Policies

The arms are P0 (sliding window, recency, verbatim), RAG (store-all, FIFO, verbatim), P1 (threshold, scored at ingest), P1-A (adaptive: score orders eviction, compress), P1-S (select: score orders eviction, drop), P2 (0.5B judge), P3 (learned, never fitted on the conversation it scores) and an oracle. Two require comment. RAG is the no-policy control: it stores everything, retrieves the top k items, and evicts FIFO when the store is full. It is not a weak baseline but the arm that a decision policy has to beat in order to have contributed anything. P0 is given the same token budget as MemGate, so the only difference between them is which turns are selected, not how many tokens may be spent.

C. One variable per comparison

The experimental design lies in the set of contrasts rather than in the set of policies. Each contrast holds everything constant except one factor:

Contrast	Held constant	Varies
P0 vs. RAG	retention, storage	read path
RAG vs. P1-S	storage, fidelity, retrieval	**what is forgotten**
P1-S vs. P1-A	storage, selection	**fidelity**

The second contrast is the one this paper depends on. RAG and P1-S hold the same verbatim turns, in the same embedding space, retrieved by the same procedure, under the same storage cap. The only difference is which turns are discarded when the cap binds: oldest-first for RAG, lowest-utility-first for P1-S. Any gap between the two is therefore attributable to the decision policy.

IV. Metrics

We report two retrieval-level metrics. Evidence recall, also called strict recall, is the fraction of questions for which every gold evidence turn reached the assembled context. Answer recall is the fraction for which the answer text survived into the context, computed over the 715 of 1,527 questions whose answer is literally recoverable from the evidence turns.

The distinction is load-bearing. Evidence recall credits an identifier: a compressed gist retains the identifier of a turn whose text it discarded, and records a hit with none of the answer behind it. In development we saw evidence recall rise from 8.1% to 14.8% while answer recall stayed at 12.8%

— the gain was entirely identifier bookkeeping. Dropping against compressing is precisely the comparison on which the two diverge, so reporting evidence recall alone would have produced the opposite conclusion. We treat answer recall as primary and report evidence recall alongside it to keep the divergence visible.

Both are forms of context recall, an upper bound on end-task accuracy rather than accuracy itself. Their advantage here is that they are deterministic and do not confound the memory policy with the reader's reasoning.

V. Experimental Setup

We use LoCoMo [15]: 10 multi-session conversations, 5,882 turns, 1,527 scoreable questions. Category-5 (adversarial) items are excluded — they carry evidence, but the correct behaviour is to decline, so retrieving it would not be success.

Retrieval and semantic supersession use all-MiniLM-L6-v2 [17] at 384 dimensions, and budgets are counted with `cl100k_base`, so a budget is an exact token count. The P2 judge is Qwen2.5-0.5B-Instruct [18], run locally; salience is read as a continuous expectation over digit-token logits in one forward pass rather than generated and parsed, which is faster, cannot fail to parse, and leaves no ties in the eviction ranking.

All intervals are paired bootstrap [19] with $B = 10{,}000$ resamples drawn at the conversation level, with an exact McNemar test [20]. The resampling unit is not a detail: 1,527 questions from 10 conversations are not 1,527 independent trials, and one of our own claims does not survive the correction (Section VI-C). Every model is a pinned local copy with the hub disabled at import, and all numeric tables are generated from the committed result files rather than transcribed.

VI. Results

A. Storage was never budgeted

Experiments in this area cap the context assembled per query and leave the store unbounded, and our own experiments did the same until this point. Under that accounting a store-everything baseline is charged nothing for holding all 419 turns of a conversation, so forgetting can only lose information. Run under those conditions, our own ablation reported the tiering and compression machinery as a net loss against simply keeping everything and retrieving, and that result was correct given how the cost was being counted.

Imposing a storage budget S changes the question from what fits in the context to what is worth keeping at all. Table I sweeps S with the context budget fixed at 2048 tokens. The ranking under the corrected accounting differs from the ranking under the original one. This is the most consequential observation in the paper, because an evaluation protocol that omits a cost answers a different question from the one it appears to ask.

B. Selection pays; compression does not

Table I reports both metrics across the sweep. P1-S, which drops the lowest-utility item when the cap binds, leads on answer recall at every budget. P1-A, which compresses that item instead of dropping it, leads on evidence recall at every budget, yet at $S = 8192$ trails even RAG on answer recall, by 3.4 points.

TABLE I
Recall against the storage budget S in tokens, with the context budget fixed at 2048. Δ is P1-S minus RAG on answer recall: identical storage, identical fidelity, identical retrieval, differing only in which turns are forgotten.

	Evidence recall (%)			Answer recall (%)			
S	RAG	P1-A	P1-S	RAG	P1-A	P1-S	Δ
1024	4.3	9.3	5.3	9.1	11.9	**15.8**	+6.7
2048	10.1	19.4	11.8	18.3	22.7	**27.8**	+9.5
4096	17.8	31.8	20.2	28.1	30.3	**37.5**	+9.4
8192	30.6	45.6	38.6	44.3	41.0	**57.8**	+13.4
16384	53.3	58.2	57.9	66.4	63.1	**71.3**	+4.9

This is the identifier problem of Section IV in practice. A demoted gist retains the evidence identifiers of a turn whose text has been discarded. An evaluation reporting only evidence recall would conclude that compression is the better use of a bounded store, and an evaluation that checks whether the answer text survived reaches the opposite conclusion.

C. Statistical significance

Table II reports each contrast with intervals resampled at the conversation level. The answer-recall gain of scored eviction over FIFO is $+9.37$ points with the interval excluding zero. The corresponding evidence-recall gain is $+2.42$ points and does not survive clustering ($p = 0.08$), so we do not claim it. We report it here rather than omitting it, since a claim that fails a correctness check is itself information about the check.

The P1-S against P1-A rows show the divergence with signs attached: $+7.13$ on answer recall and -11.53 on evidence recall, both significant. The two metrics therefore disagree in direction and not only in magnitude.

D. The rate–distortion crossover is not reached

Rate–distortion reasoning predicts compression must eventually beat selection, since a sufficiently compressed record of everything should outperform a faithful record of a subset [7]. Sweeping S downward cannot test this: every policy hits the floor first, near 0.2% evidence recall at $S = 256$. The missing axis is length, so we concatenate up to ten LoCoMo conversations — 186K tokens — holding S fixed, which raises the compression ratio to $182\times$. Table III reports no crossover: the margin of selection over compression narrows but stays positive, and the best operating point stays at the vertex, a subset at full fidelity.

One caveat. A concatenated stream is not a natural long conversation — each question concerns one constituent and the others act as distractors — so this is a stress test of retention rather than a long-dialogue benchmark. We falsify the crossover on this axis to this ratio, and claim nothing beyond.

TABLE II
PAIRED BOOTSTRAP ($B = 10{,}000$) RESAMPLED BY CONVERSATION, WITH EXACT MCNEMAR, AT $S = 4096$. THE SECOND ROW IS THE CLAIM WE DECLINE TO MAKE.

Comparison	Metric	Δ (pts)	95% CI (clustered)	McNemar p	
P1-S vs. RAG (FIFO)	answer	$\mathbf{+9.37}$	$[+5.45, +14.90]$	2.8×10^{-5}	✓
P1-S vs. RAG (FIFO)	evidence	$+2.42$	$[-1.48, +6.43]$	0.08	N.S.
P1-S vs. P1-A (compress)	answer	$\mathbf{+7.13}$	$[+4.32, +10.28]$	2.1×10^{-5}	✓
P1-S vs. P1-A (compress)	evidence	$\mathbf{-11.53}$	$[-14.73, -8.47]$	1.8×10^{-23}	✓
P1-S vs. P0 (recency)	answer	$\mathbf{+19.16}$	$[+15.42, +24.13]$	8.2×10^{-19}	✓
P1-S vs. P0 (recency)	evidence	$\mathbf{+10.15}$	$[+6.58, +13.84]$	1.2×10^{-15}	✓

TABLE III
ANSWER RECALL AGAINST COMPRESSION RATIO AT $S = 2048$. THE RIGHTMOST COLUMN IS THE MARGIN OF DROPPING OVER COMPRESSING.

Ratio	RAG	P1-A	P1-S	P1-S − P1-A
9×	18.3	22.7	**27.8**	+5.2
18×	10.8	13.0	**18.9**	+5.9
45×	7.8	10.3	**11.6**	+1.3
91×	4.6	7.4	**8.7**	+1.3

TABLE IV
ANSWER RECALL (%) WITH THE BUDGET DENOMINATED IN BYTES, CHARGING 1,536 B PER RETRIEVABLE ITEM FOR ITS EMBEDDING. P0 MAINTAINS NO INDEX AND IS THEREFORE FLAT.

Budget (KiB)	P0	RAG	P1-A	P1-S
32	**18.3**	5.0	5.9	15.1
64	18.3	11.0	9.4	**24.9**
128	18.3	20.0	11.9	**33.1**
256	18.3	33.0	15.8	**46.0**
512	18.3	50.3	22.2	**61.0**

E. The embedding index is not free

Denominating storage in text tokens leaves out the index that makes retrieval possible. A 384-dimensional `float32` vector occupies 1,536 B, while a LoCoMo turn averages about 130 B of text. The index therefore outweighs the content it indexes by roughly a factor of 12, and the limiting resource becomes the number of stored items rather than their length.

Table IV repeats the sweep with the budget denominated in bytes and each retrievable item charged for its vector. Two results follow. First, the selection finding of Section VI-B is robust to the change of units, since P1-S leads at every budget above 32 KiB. Second, below approximately 40 KiB an embedding index cannot pay for its own storage: P0, which maintains no index, performs best. A protocol that counts only text cannot observe this regime, which is the regime that a memory layer on a constrained device would occupy.

F. Scorer analysis

If the decision is where the value lies, then the scorer that makes it is the component to improve. Table V removes one mechanism at a time. Retrieval dominates, at -31.6 evidence points, and scored eviction is worth 6.4 points. The demotion row is marked because it shows the same divergence in miniature: removing demotion costs 11.5 evidence points and gains 7.1 answer points.

TABLE V
COMPONENT ABLATION AT $S = 4096$, SHOWING THE CHANGE FROM THE FULL SYSTEM. †: THE ROW ON WHICH THE TWO METRICS DISAGREE IN DIRECTION.

Component removed	Δ evidence	Δ answer
Retrieval (Tier 3)	-31.6	-28.0
Demotion on pressure †	-11.5	$+7.1$
Scored eviction (→ FIFO)	-6.4	-7.4
Informative compressor	-4.4	-1.7
Consolidation merge	-3.4	-4.8
Semantic supersession	-0.3	-0.3

Table VI varies only the scorer, everything else fixed at the configuration of Section VI-B. Three results follow, all significant under clustering. Scoring is worth doing: the hand-written heuristic gains $+11.19$ answer points over unscored recency. The LLM judge nonetheless loses: in the pre-fix run, Qwen2.5-0.5B-Instruct reaches 30.9% against the heuristic's 36.8%, -5.87 points with the interval excluding zero, at some fifteen minutes of GPU time per corpus against microseconds for a regular expression. And the headroom is real: a logistic scorer fitted under leave-one-conversation-out reaches 48.3%, $+11.47$ over the heuristic, so the judge was not limited by a ceiling that does not exist.

We attribute the judge's failure to what it is shown. It rates each turn in isolation, and salience is not a property of a turn considered alone: a phone number matters because a later turn asks for it. The fitted scorer sees the same isolated text but absorbs corpus-level base rates a per-turn prompt cannot convey — its strongest single feature is turn length, ahead of every hand-designed cue. Under leave-one-conversation-out it never sees the questions of the conversation it scores, so it is deployable in the way any supervised component is: fitted offline on labelled conversations, then applied to new ones. What this does not establish is that the base rates it absorbs carry over to a different kind of conversation, which Section VI-G tests.

G. A learned scorer across two corpora

LoCoMo's dialogues are generated by an LLM, so a scorer fitted to them may learn the generator's style rather than

TABLE VI
SINGLE-COMPONENT SCORER SWAP AT $S = 4096$ WITH CONTEXT 2048. POLICY, RETRIEVAL AND COMPRESSION ARE FIXED AND ONLY THE SCORER VARIES. P3 IS FITTED ON THE OTHER NINE CONVERSATIONS. PRE-FIX RUN (SECTION VIII).

	What the scorer sees	Evid.	Ans.	Δ vs. P1	95% CI
P0	position only	12.6	25.6	−11.19	[−17.24, −6.19]
P1	regex + shallow NER	19.8	**36.8**	—	—
P2	the turn, in isolation	14.0	30.9	−5.87	[−10.77, −1.76]
P3	turn + fitted base rates	33.4	48.3	+11.47	[+4.90, +17.86]

what people later ask about. We test this on REALTALK [16]: ten conversations between real people over 21 days, 8,944 turns and 660 questions with turn-level evidence in LoCoMo's schema, of which 160 are answer-recoverable. The scorer is a logistic regression over 15 shallow features of the message (length, digits, dates, capitalisation, question and filler cues) concatenated with its 384-dimensional MiniLM embedding, labelled by whether any question cites the turn as evidence. Every held-out conversation is scored by three models that never saw it: trained on the other conversations of the same corpus, on the other corpus only, or on all 19 other conversations pooled. Policy, budgets and retrieval are those of Section VI-B with the fixed heuristic (Section VIII); only the scorer varies.

Table VII reports the result. On LoCoMo the pooled scorer reaches 52.0% answer recall against the heuristic's 37.5%, a gain of $+14.55$ points, and pooling beats same-corpus training by $+3.36$ ($[+0.27, +6.52]$), so adding a different corpus helped rather than diluted. Transfer works in one direction only. Trained on REALTALK alone, the scorer still beats the heuristic on LoCoMo by $+9.09$; trained on LoCoMo alone, it falls below the heuristic on REALTALK (-4.38, n.s.) and is significantly worse than same-corpus training there (-10.62, $[-18.31, -3.45]$). At this budget no gain over the heuristic on REALTALK is significant, which with 160 questions in ten clusters is a statement about power as much as about the scorer.

$S = 4096$ is the budget of the headline comparison, not one chosen for this scorer, so Table VIII repeats the pooled run at four other budgets. On LoCoMo the gain is significant from $S = 2048$ to 16384 and largest at 4096; it shrinks at both ends, where the store holds too little for any ranking to matter or nearly everything anyway. On REALTALK the gain is significant at $S = 8192$ ($+16.88$, $[+6.31, +26.72]$) and 16384. The one budget at which neither corpus shows a significant gain is $S = 1024$, about 5% of a conversation.

The embedding is what carries across. With the hand features alone, the pooled scorer is significantly *worse* than the heuristic on REALTALK (-8.75, $[-17.45, -2.27]$): its feature weights encode LoCoMo's style, such as long turns ending in a period, which people messaging each other do not write. This is the same failure the heuristic shows on LongMemEval (Section IX). Logistic regression was kept after comparing it with boosted trees and a 64-unit MLP on the same folds: neither beat it by more than the spread between held-out conversations, and the trees were much slower both to fit and on the write path.

TABLE VII
LEARNED SCORER ON HELD-OUT CONVERSATIONS OF TWO CORPORA, HAND FEATURES PLUS MINILM, $S = 4096$, CONTEXT 2048. POLICY P1-S; ONLY THE SCORER VARIES. Δ IS ANSWER RECALL AGAINST P1, WITH 95% INTERVALS RESAMPLED BY CONVERSATION; BOLD EXCLUDES ZERO.

Scorer (trained on)	AUC	Evid.	Ans.	Δ vs. P1	95% CI
LoCoMo (715 answer-recoverable questions)					
RAG (FIFO, no scorer)	–	17.8	28.1	—	—
P1 heuristic	0.599	20.2	37.5	—	—
P3, same corpus	0.793	32.9	48.7	**+11.19**	[+4.72, +17.06]
P3, other corpus only	0.692	30.1	46.6	**+9.09**	[+4.53, +13.46]
P3, pooled (shipped)	0.791	34.0	52.0	**+14.55**	[+10.40, +19.70]
REALTALK (160 answer-recoverable questions)					
RAG (FIFO, no scorer)	–	10.5	16.9	—	—
P1 heuristic	0.624	12.6	32.5	—	—
P3, same corpus	0.737	18.5	38.8	+6.25	[−3.01, +13.42]
P3, other corpus only	0.681	11.2	28.1	−4.38	[−12.50, 0.00]
P3, pooled (shipped)	0.749	17.3	36.2	+3.75	[−2.92, +8.65]

TABLE VIII
THE POOLED LEARNED SCORER (P3) AGAINST THE P1 HEURISTIC ACROSS THE STORE BUDGET S, ANSWER RECALL (%), CONTEXT 2048, SAME PROTOCOL AS TABLE VII. Δ IS P3 MINUS P1 WITH 95% INTERVALS RESAMPLED BY CONVERSATION; BOLD EXCLUDES ZERO.

	LoCoMo			REALTALK		
S	P1	P3	Δ [95% CI]	P1	P3	Δ [95% CI]
1024	15.8	19.0	+3.22 [0.00, +6.40]	8.1	10.0	+1.88 [−1.95, +5.36]
2048	27.8	33.7	**+5.87** [+1.83, +9.88]	16.2	15.0	−1.25 [−10.62, +3.98]
4096	37.5	52.0	**+14.55** [+10.40, +19.70]	32.5	36.2	+3.75 [−2.92, +8.65]
8192	57.8	69.1	**+11.33** [+6.06, +16.41]	41.2	58.1	**+16.88** [+6.31, +26.72]
16384	71.3	77.2	**+5.87** [+2.04, +10.56]	61.3	68.1	**+6.88** [+2.12, +12.21]

H. End-task accuracy

Every result above is context recall, which bounds end-task accuracy without measuring it. We therefore send each assembled context through one fixed reader, Qwen2.5-1.5B-Instruct-4bit decoding greedily for at most 32 tokens, and score the output with the same normaliser the answer-recall metric applies to the context. "The answer reached the context" and "the answer reached the output" are then one predicate evaluated at two points in the pipeline, so the ceiling comparison is exact rather than approximate. The run covers all 715 answer-recoverable questions with two controls: a closed-book arm that receives no memory at all, and an oracle arm that receives exactly the cited evidence.

Table IX reports the outcome. The closed-book floor is 1.7%, so parametric knowledge contributes almost nothing on this benchmark and the margin above that floor is attributable to the memory layer. P1-S reaches 18.5% against RAG's 12.4%. Resampling by conversation as before, the realised gap is $+6.0$ points with a 95% interval of $[+3.3, +9.1]$, which is 69% of the $+8.7$-point recall gap between the same two arms in this pre-fix run (Section VIII).

Accuracy is not bounded by answer recall, since a reader may guess, so the table separates the two regimes. Accuracy given that the answer was present is 49.4% for P1-S against 42.3% for RAG, while accuracy given that it was absent is

TABLE IX
END-TASK ACCURACY THROUGH ONE FIXED READER AT $S = 4096$ WITH CONTEXT 2048, OVER ALL 715 ANSWER-RECOVERABLE QUESTIONS. ACC.|PRESENT IS ACCURACY RESTRICTED TO THE QUESTIONS WHOSE ANSWER REACHED THE CONTEXT; IT IS UNDEFINED FOR THE ARMS THAT NEVER PLACE ONE. PRE-FIX RUN (SECTION VIII).

Arm	Acc.	F1	Ans. present	Acc.|present
Closed-book *(floor)*	1.7	5.0	0.0	–
P0 sliding window	8.3	11.9	18.3	40.5
RAG store-all	12.4	17.7	28.1	42.3
P1-A adaptive (compress)	13.1	19.9	30.1	42.3
P1-S select (drop)	**18.5**	24.7	36.8	49.4
Oracle *(ceiling)*	67.3	75.7	100.0	67.3

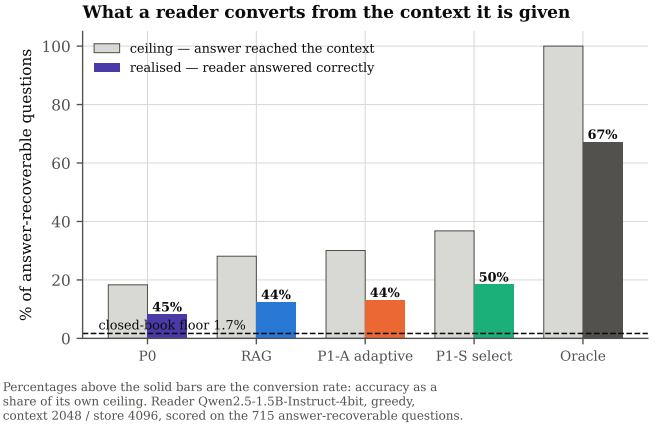


Fig. 1. Reader accuracy by memory policy, with the closed-book floor and the oracle ceiling. The ordering matches the recall ordering, at a lower level.

below 1% for every arm. The selection advantage therefore arrives through both available routes: P1-S places the answer in context more often, and the contexts it assembles are also the ones the reader uses more successfully. The oracle arm converts a complete context into a correct answer 67.3% of the time, which bounds what any memory policy can deliver through this reader and locates the remaining headroom in the reader rather than in the memory layer.

VII. FROM HARNESS TO APPLICATION

The policy is also a working system. A chat application wraps a reader model in the same `observe()` and `build_context()` calls the experiments use: every message, the user's and the model's reply alike, is scored and routed when it arrives, and each question is answered from the context MemGate assembles, never from the raw history. The default configuration runs offline on an 8 GB laptop with Qwen2.5-1.5B-Instruct as the reader; a hosted model can be substituted, or none, in which case the application shows the context the reader would have received. The scorer is the pooled model of Section VI-G, refitted on all 20 conversations and shipped as 399 coefficients in plain JSON; scoring a message with it takes about 0.25 ms once the embedding is computed, and if the MiniLM embedder is unavailable it falls back to the heuristic rather than scoring with the wrong features.

Two properties follow from reusing the experimental code. First, persistence is exact: a conversation is saved as its messages and restored by replaying them through `observe()`, which is deterministic, so the tiers after a restart are the tiers before it. Second, the application can check itself: a built-in benchmark run recomputes the RAG and P1-S arms on all of LoCoMo at $S = 4096$ on a CPU and compares them with the committed results, reproducing 28.1% and 37.5% exactly. A side panel shows each message's utility u, the tier it was routed to or the reason it was forgotten, and the exact context sent to the reader, so every decision the paper measures in aggregate can be inspected one message at a time. The application runs at the experimental budgets by default and offers a reduced scale (Tier 1 of four messages, a 512-token context and $S = 150$) under which eviction becomes visible within a few messages; these sizes are the only settings that differ from the experiments.

VIII. THREATS TO VALIDITY

Three errors changed our results before we detected them. Each was caught by an invariant rather than by reading the code.

A policy that read its evaluation label. LoCoMo supplies a `fact_id` per evidence turn, and an early consolidation rule promoted items using it — the policy was consulting the answer key. The effect was worth 45 to 47 recall points and invalidated our first reported result. The remedy is an invariant, not a convention: strip every evaluation label from the input and the resulting store must be bit-identical. Each decision point added since was placed under it before any number was reported.

Component value depends on the current bottleneck. We first measured the sentence embedder at $+0.2$ points and concluded it barely mattered. That held only in the regime measured, where retrieval was saturated against a write path discarding 93% of the conversation; once that bottleneck was removed the same substitution was worth $+19.7$. Ablations run against a bottlenecked pipeline systematically undervalue whatever would matter after the dominant term moves.

A failure invisible in the aggregate. Batched judge scoring returned `NaN` for short turns specifically. Precision was the natural hypothesis and `float32` did not help; the cause was left-padding, producing fully masked attention rows. Without an explicit guard every filler turn would have scored `NaN` and inverted the eviction order.

A fourth defect was found after the main runs: the heuristic scored each turn with its speaker tag, which blocked the filler rule and gave nearly every turn the proper-noun bonus (5,863 of 5,882 LoCoMo turns were labelled facts). Recall tables and Table VII use the fixed scorer; the judge, end-task, abstractive and LongMemEval runs predate it. The fix moves the headline P1-S result from 36.8% to 37.5%, and it shrinks the learned scorer's lead over the heuristic at $S = 2048$ from $+5.2$ to $+2.7$

points, because part of that lead was the defect handicapping the heuristic.

IX. Limitations

Few conversations. Clustering by conversation is the correct unit and leaves few clusters, so intervals are wide; we report the effects that do not survive as not significant. REALTALK adds ten conversations but only 160 answer-recoverable questions, too few to resolve most of its differences.

A single reader model. Section VI-H uses one 1.5B four-bit model decoding greedily. We claim the ordering of policies, not the absolute accuracies: the oracle arm's 67.3% shows much of the remaining error is the reader, and a larger one would raise every arm and could compress or widen the margins.

Compression was tested in both forms. Replacing P1-A's word-selection with a 1.5B model rewriting each evicted turn is worth $+4.06$ answer points (CI $[+0.84, +6.66]$), so part of the negative result was the compressor rather than compression. Selection still wins (pre-fix run): 36.8% against 34.1%, at 2.1 s against 3,892 s and 4,879 generations. The claim narrows to: a better compressor helps, and does not overtake dropping.

Concatenated streams are synthetic, as discussed in Section VI-D.

The scorer is tuned to this corpus. On a 94-item LongMemEval [3] subsample, whose turns average 210 tokens against LoCoMo's 32, the heuristic's separation between evidence and non-evidence turns reversed in sign and scored eviction lost to FIFO. Read the $+9.37$ of Section VI-B as what a policy recovers *given a scorer suited to the corpus*, not as a property of scored selection in general. The learned scorer of Section VI-G has been tested on LoCoMo and REALTALK only, and LongMemEval, where the heuristic's sign reversed, remains untested with it. Its labels are distant supervision (a turn is positive if any question cites it), and that supervision is twice as dense on LoCoMo (24% of turns) as on REALTALK (12%).

Single embedder and single judge. The negative result of Section VI-F is evidence about a 0.5B model prompted per turn, not about LLM-judged salience in general. A larger judge, or one shown the neighbouring turns, remains untested and is the most direct way to challenge the finding.

X. Conclusion

Under the accounting normally used here, tiered compression is a net loss against storing everything and retrieving — but that is an artefact of leaving the store unbounded. Bound it, and the decision of which items to forget is worth $+9.37$ answer-recall points at identical storage, fidelity and retrieval, while compressing an evicted item rather than dropping it loses 7.13. The crossover at which compression should begin to pay is not reached to $182\times$. Charging the budget for the embedding index reverses the ranking below about 40 KiB. An LLM judge, the obvious route to a better scorer, loses to a regular expression, for an identifiable reason: salience is not a property of a turn in isolation. A logistic scorer over a sentence embedding, trained on LLM-generated and human conversations together, beats the heuristic by $+14.55$ points on held-out LoCoMo conversations, and by a significant margin at every store budget from 2048 to 16384 tokens; it transfers from human dialogue to generated dialogue but not the other way. Through a fixed reader, $+6.0$ of the pre-fix $+8.7$ points survive as correct answers.

The practical recommendation is narrow. Retain a scored subset at full fidelity rather than compressing everything, and denominate the budget in bytes with the index included. The methodological recommendation is broader. Three of the results above changed in sign or in magnitude once a measurement error was removed, and none of those errors was visible in the aggregate numbers they affected.

Two questions remain. The pooled scorer's gains on human conversation are significant only once the store holds about 40% of a conversation, and it is untested on LongMemEval, where hand-crafted cues inverted; more human-written evidence, rather than a larger model, is what would settle both. And a scorer learned online from retrieval feedback is already statistically level with the heuristic on LoCoMo without labels or future questions (-2.94, n.s.); whether it can be seeded with the pooled scorer and then adapt to each user is the step from a scorer fitted to benchmarks to one fitted to the person using the application.

Acknowledgment

The authors thank the faculty of SVKM's NMIMS, Indore Campus, for their guidance throughout this capstone project.

References

[1] Chroma Research, "Context rot: How increasing input tokens impacts LLM performance," Chroma, Tech. Rep., 2025.

[2] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the middle: How language models use long contexts," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 157–173, 2024.

[3] D. Wu, H. Wang, W. Yu, Y. Zhang, K.-W. Chang, and D. Yu, "LongMemEval: Benchmarking chat assistants on long-term interactive memory," *arXiv preprint arXiv:2410.10813*, 2024.

[4] C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, and J. E. Gonzalez, "MemGPT: Towards LLMs as operating systems," *arXiv preprint arXiv:2310.08560*, 2023.

[5] W. Zhong, L. Guo, Q. Gao, H. Ye, and Y. Wang, "MemoryBank: Enhancing large language models with long-term memory," in *Proc. AAAI Conf. Artificial Intelligence*, vol. 38, no. 17, 2024, pp. 19 724–19 731.

[6] Z. Zhang, X. Bo, C. Ma, R. Li, X. Chen, Q. Dai, J. Zhu, Z. Dong, and J.-R. Wen, "A survey on the memory mechanism of large language model based agents," *arXiv preprint arXiv:2404.13501*, 2024.

[7] R. Colaco and A. Lahjouji, "A rate–distortion framework for memory in language agents," *arXiv preprint arXiv:2607.08032*, 2026.

[8] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proc. ACM Symp. User Interface Software and Technology (UIST)*, 2023.

[9] W. Xu, Z. Liang, K. Mei, H. Gao, J. Tan, and Y. Zhang, "A-MEM: Agentic memory for LLM agents," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[10] H. Jiang, Q. Wu, C.-Y. Lin, Y. Yang, and L. Qiu, "LLMLingua: Compressing prompts for accelerated inference of large language models," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2023, pp. 13 358–13 376.

[11] J. Mu, X. L. Li, and N. Goodman, "Learning to compress prompts with gist tokens," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
[12] A. Chevalier, A. Wettig, A. Ajith, and D. Chen, "Adapting language models to compress contexts," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
[13] G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis, "Efficient streaming language models with attention sinks," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2024.
[14] Z. Zhang, Y. Sheng, T. Zhou, T. Chen, L. Zheng, R. Cai, Z. Song, Y. Tian, C. Ré, C. Barrett, Z. Wang, and B. Chen, "H2O: Heavy-hitter oracle for efficient generative inference of large language models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
[15] A. Maharana, D.-H. Lee, S. Tulyakov, M. Bansal, F. Barbieri, and Y. Fang, "Evaluating very long-term conversational memory of LLM agents," in *Proc. Annu. Meeting Assoc. Computational Linguistics (ACL)*, 2024.
[16] D.-H. Lee, A. Maharana, J. Pujara, X. Ren, and F. Barbieri, "REALTALK: A 21-day real-world dataset for long-term conversation," *arXiv preprint arXiv:2502.13270*, 2025.
[17] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
[18] Qwen Team, "Qwen2.5 technical report," *arXiv preprint arXiv:2412.15115*, 2025.
[19] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Chapman & Hall, 1993.
[20] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.